

Emanuele Rossi

[✉ emanuele.rossi1909@gmail.com](mailto:emanuele.rossi1909@gmail.com) [🌐 Website](#) [🎓 Scholar](#) [in LinkedIn](#) [🐙 GitHub](#)

Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously social graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Google Scholar metrics: 4,261 citations; h-index 15 (July 2026).

Education

Imperial College London Oct 2020 – Feb 2024

Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein

- Research focused on **graph machine learning methods** for large real-world systems

University of Cambridge Oct 2018 – Jun 2019

Master in Advanced Computer Science, Distinction (4.0 GPA)

Imperial College London Oct 2015 – Jun 2018

Bachelor in Computer Science, First Class Honors (4.0 GPA)

Experience

Postdoctoral Researcher Rome (remote)

GLADIA Research Group [🔗](#), Sapienza University of Rome – With Prof. E. Rodolà Oct 2025 – Present

- Developing machine learning methods for animal communication.

Machine Learning Researcher NYC (remote)

VantAI Jan 2024 – Aug 2025

- **Co-led development of Neo-1** [🔗](#), a state-of-the-art all-atom latent diffusion model for multi-modal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.

Machine Learning Researcher London

Twitter Jun 2019 – Feb 2023

- Developed **graph ML models** for Twitter's social graph and deployed a recommendation model that generated **1.2M additional daily follows**

Machine Learning Researcher London

Fabula AI Mar 2019 – Jun 2019

- Developed graph ML models for fake news detection as part of Fabula AI's small research team before [Twitter acquired the company](#) [🔗](#)

Software Engineering Intern California

Google Jul 2017 – Oct 2017

Selected Publications

[Full list](#)

Animal Communication & Audio

- *F. Mustun et al.* OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review 2026*.
- *E. Rossi and E. Rodolà.* [Communicating Sound Through Natural Language](#) [🔗](#). *arXiv 2026*.
- *D. Marincione et al.* [Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models](#) [🔗](#). *NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%)*.

Graph Learning

- *E. Rossi et al.* [Temporal Graph Networks for Deep Learning on Dynamic Graphs](#) [🔗](#). *ICML GRL Workshop 2020*. **1,700+ citations** and **1,000+ GitHub stars** [🔗](#).

- *E. Rossi**, *F. Frasca** et al. [SIGN: Scalable Inception Graph Neural Networks](#) [↗](#). *ICML GRL Workshop 2020*. **600+** citations and used in [production at Airbnb](#) [↗](#).
- *B. Chamberlain*, *S. Shirobokov*, *E. Rossi* et al. [Graph Neural Networks for Link Prediction with Subgraph Sketching](#) [↗](#). *ICLR 2023, Oral (top 5%)*. **200+** citations.

Leadership

Co-Founder, LeadTheFuture ([Forbes article](#) [↗](#))

Jun 2018 – Mar 2023

- Built a non-profit mentoring community with more than **200 mentors** and **500 selected mentees**